

Question Paper

Exam Date & Time: 24-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Semiconductor Device Theory and Fabrication [ECE_5152]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

1) Explain the Czochralski Method of Crystal Growth (5)

A) Course Outcome: "CO1" / Cognitive Level: "Understand"

B) Calculate the maximum pull rate for a 12" diameter CZ crystal. ($L = 430 \text{ cal gm}^{-1}$, $N = 2.328 \text{ gm/cm}^3$, $\sigma = 5.67 \times 10^{-5} \text{ erg/ (cm}^2 \text{ sec K}^4)$ $\times 2.39 \times 10^{-8} \text{ cal/erg}$, $\epsilon = 0.55$, $T_M = 1690 \text{ K}$, $k_M = 0.048 \text{ cal/(sec cm K)}$) (5)

Course Outcome: "CO1" / Cognitive Level: "Application"

2) Differentiate between gate oxide and field oxide (2)

A) Course Outcome: "CO1" / Cognitive Level: "Application"

B) Explain the masking properties of silicon oxide in IC Fabrication (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

C) A bare silicon wafer is oxidized in dry O_2 at 1200 °C for 2 hours. (5)

Given: $A = 0.040$, $B = 0.045$, $\tau = 0.027$, calculate oxide thickness and Si thickness consumed.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

3) Explain Intrinsic and Extrinsic diffusion in semiconductors with the help of a neat graph showing the variation of diffusion coefficient (4)

A) $D/D_i(T)$ with carrier concentration ratio $n/n_i(T)$

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) State Fick's first law of diffusion and explain the significance of the negative sign in Fick's first law. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) **A pre-deposition process is carried out for 15 minutes on an n -type silicon wafer with a bulk doping concentration of 10^{17} atoms/cm³ at 950°C using diborane gas. Determine the p - n junction depth given that the surface solubility of boron at 950°C is 3.8×10^{20} atoms/cm³, the intrinsic diffusivity (D_0) is $0.76 \text{ cm}^2/\text{s}$, and the Arrhenius activation energy (E_a) is 3.46 eV.** (4)

$$(\text{erfc}-1(2.6316 \times 10^{-4}) \approx 2.5803.)$$

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4) Explain the steps involved in pattern transfer (4)

- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Draw the exposure response curve of both the positive and negative photoresists (2)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- C) Describe the exposure methods of photolithography (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) A 100 nm thick oxide layer is etched using a wet chemical etchant. After 10 minutes of etching, the remaining thickness is 30 nm. (5)

- A) Find: (a) The etch rate in nm/min. (b) The time needed to completely etch the oxide layer.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Compare both the FINFET and GAA advanced transistor structures with neat diagrams. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM PG-2025-2026	ECE_5152:Semiconductor Device Theory and Fabrication

#	Course Outcome	Description	Marks
1	1	CO 1	12
2	2	CO 2	12
3	3	CO 3	12
4	4	CO 4	14
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM PG-2025-	ECE_5152:Semiconductor Device Theory and

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	7	25(50%)
4	Analysis	5	20(40%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	13	50 (100%)